

# UNICOM TIC MANAGEMENT SYSTEM

## C# PROJECT REPORT

### Default Login Credentials

- Username: admin
- Password: admin123@

### Key Features Implemented

- User ID and Password Management
- Admin Management
- Staff Management
- Lecturer Management
- Course and subject Management (\*.....\*)
- Lecturer Management with course subject (\*.....\*)
- Student Management with Batch
- Exam and Marks Management

### Technologies Used

- Database: SQLite
- Architecture: MVC (Model-View-Controller)
- Platform: Windows Forms (C#)
- Programming Paradigm: Object-Oriented Programming (OOP)

### Rules for using this application

There are rules and regulations for the system to do its optimum work .

When you are a new user newly started to use it .It has a default username and password as i mention above in the start .After entering into the main page all the features for the admin will be displayed. You need to register an admin first, then you need to register the staff after that u should create a batch, then you should create the course after that the lecturer should be created first before creating the subject. Subjects can be allocated only when the teacher is available else message box appears to create the lecturer.

After the subject is created you should create a course\_subject table which handles many to many relationship between subjects and courses.After that you have to select which teacher is going to teach which course according to the eligible subject.

After all this specifications only a student can be selected and the respective exam and marks can be added to the students according to the batch. Search options are available for the main roles. Every single delete for the main Admin, Staff, Lecturer and Students are soft deletes. After the course is completed, you only need to delete the course and it will automatically delete the students on the student table.

Time Table

Exam

Marks

Student

Course\_Subject

Lecturer

Subject

Course

Lecturer Course Subject

Staff

Room

Batch

Admin

Log out

Unicom TIC Management System



**Unicom TIC**  
Training & Innovation Centre  
Skills Today, Success Tomorrow

Select First

Add

Search

Name

Kathir

Special In

C#

Address

inuvil

User Name

kathir123@|

Username available

NIC

75497649674

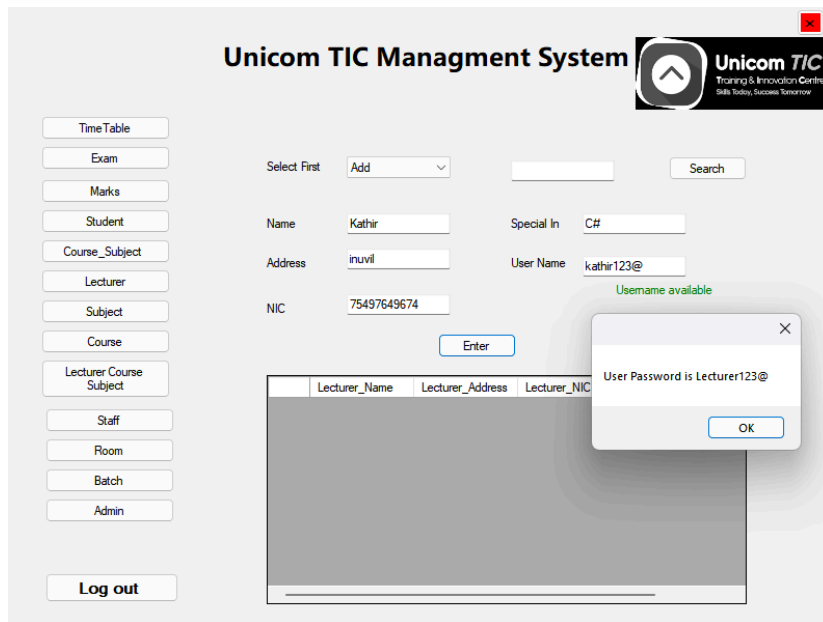
Enter

| Lecturer_Name | Lecturer_Address | Lecturer_NIC | Special_In | Us |
|---------------|------------------|--------------|------------|----|
|---------------|------------------|--------------|------------|----|

This is Lecturer creation

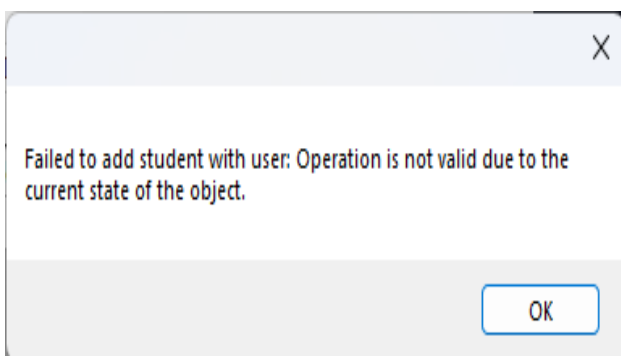
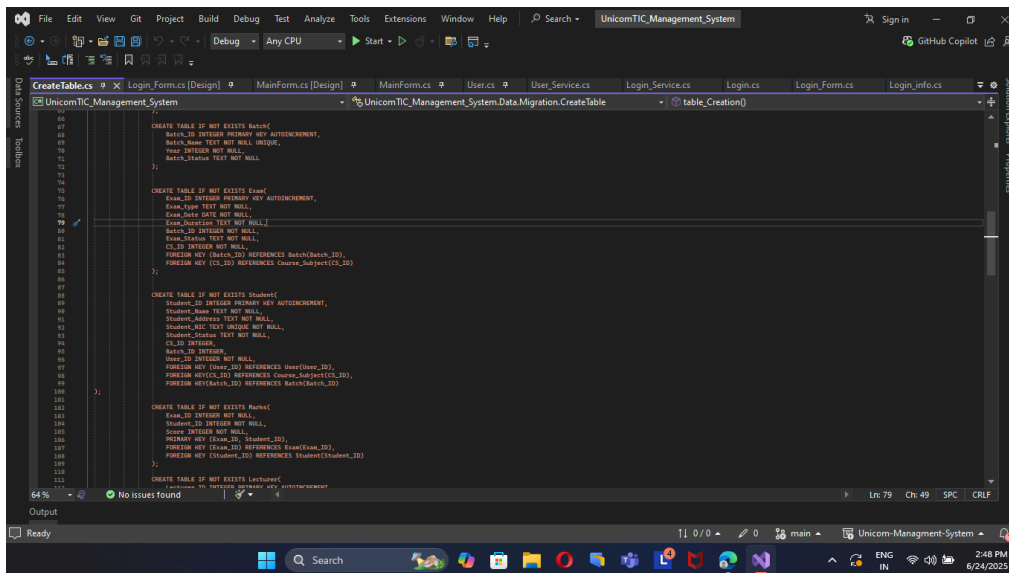
The screenshot displays the 'Unicom TIC Management System' interface. On the left, a vertical sidebar contains buttons for 'TimeTable', 'Exam', 'Marks', 'Student', 'Course\_Subject', 'Lecturer', 'Subject', 'Course', 'Lecturer Course Subject', 'Staff', 'Room', 'Batch', 'Admin', and 'Log out'. The main area features a form for creating a new lecturer. The form includes a 'Select First' dropdown menu set to 'Add', a 'Search' button, and input fields for 'Name' (Kathir), 'Special In' (C#), 'Address' (inuivil), 'User Name' (kathir123@), and 'NIC' (75497649674). A green message 'Username available' is displayed below the 'User Name' field. An 'Enter' button is located below the 'NIC' field. A dialog box with the title 'Lecturer 'Kathir' added successfully.' and an 'OK' button is overlaid on the form. Below the form, a table with columns 'Lecturer\_Name', 'Lecturer\_Address', and 'Lecturer\_N' is visible, though its content is obscured by the dialog box.

Successful Creation message username are unique to every single user and its is checked simultaneously when typing and gives error messages when needed you cannot change the username ones created .



Default userpassword

This is a sample table creation query used here .



Relevant error messages .

## Challenges Faced and Solutions

- The application was initially intended to be a large-scale system, but due to limited time and resources, the development had to be scaled down and completed quickly. This required prioritizing key features and managing the scope effectively to meet deadlines.
- Designing a flexible database structure using SQLite was a challenge, especially when handling relational data between students, teachers, courses, and exam records. Careful planning and normalization helped maintain consistency.
- Implementing user authentication and role-based access control was complex, especially in managing multiple user types (Admin, Teacher, Staff). To address this, a simplified login system with encrypted credentials and validation was used.
- One of the key challenges was managing form destruction and memory leaks in Windows Forms. Some forms were not properly disposed of, leading to performance issues. This was resolved by ensuring all opened forms were closed correctly and resources were released.
- Ensuring real-time data reflection in the UI after updates (e.g., adding a student or updating marks) required efficient use of data bindings and refresh methods, which took additional debugging and testing.